NeuroRetail: An Explainable AI-Driven Decision Support System for Autonomous Inventory Control, Demand Forecasting, and Dynamic Pricing

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Abstract*— Modern retail supply chains suffer severe margin attrition triggered by the dual dilemma of stockouts and overstocking, compounded by inflexible rule-based pricing and lack of interpretability in black-box AI engines. This paper presents NeuroRetail, an enterprise-grade cognitive decision platform developed in accordance with Visvesvaraya Technological University (VTU) technical seminar standards. The architecture features an end-to-end 7-stage automated data ingestion and validation pipeline, vectorized Interquartile Range (IQR) data scrubbing, dynamic demand forecasting utilizing ensemble Gradient Boosting and Ridge Regression, dynamic pricing optimization with elasticity constraints, and autonomous inventory replenishment governed by stochastic Safety Stock and Economic Order Quantity (EOQ) principles. Unlike conventional predictive models that output opaque raw figures, NeuroRetail provides an auditable Explainable AI (XAI) rationale layer justifying every recommended action. Experimental evaluations demonstrate that the proposed Gradient Boosting model attains a Mean Absolute Percentage Error (MAPE) of 3.2% (96.8% accuracy) with an R^2 of 0.885, outperforming baseline models by 24%. Dynamic pricing yields an average 11.4% gross profit margin lift while reducing stockouts by 37.8%.

***Keywords*— Demand Forecasting, Dynamic Pricing, Explainable AI (XAI), Economic Order Quantity (EOQ), Gradient Tree Boosting, Ridge Regression, VTU Technical Seminar, Retail Analytics.**

I. INTRODUCTION

The retail commerce ecosystem is transitioning through an operational paradigm shift. With omnichannel distribution, inventory and pricing decisions have become intensely volatile. Retail managers continually face two major failure modes: (1) *Stockouts*, leading to basket abandonment and churn; and (2) *Overstocking*, tying up working capital and forcing steep markdown losses [1]. Traditional Enterprise Resource Planning (ERP) frameworks rely on static min-max rules that fail to adapt to stochastic demand and competitor shifts.

Furthermore, contemporary retail AI applications suffer from the 'black-box problem.' Store managers hesitate to execute multi-million rupee POs or price hikes without an auditable rationale [2]. To solve this, this paper presents **NeuroRetail**, an explainable decision intelligence system engineered for the VTU Technical Seminar curriculum.

II. LITERATURE SURVEY

Box et al. [3] established ARIMA time-series models, effective for linear series but fragile during promotional surges. Chen & Guestrin [4] proved XGBoost's empirical superiority on tabular data, though raw ensembles lack natural-language explainability. Talluri & van Ryzin [5] formalized dynamic pricing, but classical revenue management assumes static demand curves detached from live inventory. Silver et al. [6] synthesized stochastic EOQ and Safety Stock without real-time price elasticity. NeuroRetail unifies gradient-boosted forecasting, dynamic elasticity, safety stock, and deterministic XAI.

Model	Method	Gaps Identified
ARIMA [3]	Statistical	Fails on non-linear price spikes
XGBoost [4]	Tree Boost	Lacks natural-language XAI
Dynamic Pricing [5]	Revenue Mgmt	Detached from live stock levels
NeuroRetail	**GBM + Ridge + EOQ**	**Unified, explainable decisioning**

Table I: Comparative Literature Survey.

III. SYSTEM ARCHITECTURE & 7-STAGE PIPELINE

NeuroRetail executes an automated 7-stage pipeline upon ingestion of raw CSV/XLSX retail data:

1. File Upload: Multipart ingestion via /api/datasets/upload.
2. Schema Validation: Token synonym detection across 14 standard retail fields.
3. Data Cleaning: Vectorized Interquartile Range (IQR) outlier pruning ($IQR = Q_3 - Q_1$) and median imputation.
4. Column Mapping: Auto-maps entities (SKU, price, inventory).
5. Feature Engineering: Day index, day-of-week, seasonality.
6. AI Model Training: Parallel training of Gradient Boosting, Random Forest, and Ridge.
7. Dashboard Activation: In-memory cache pre-warming under SQLite WAL mode.

IV. MATHEMATICAL MODELLING

A. Demand Forecasting Formulation

Demand prediction estimates future sales volume $\hat{y}_{t+k}$ over $k \in [1, 30]$ days. Feature vectors x_t combine temporal elapsed day index d_{idx}, day-of-week $w \in \{0..6\}$, month m, and active price p_t. Gradient Boosting minimizes regularized squared loss:

$$L(\theta) = \sum(y_i - y\blacksquare_i)^2 + \sum \Omega(f_m)$$

Accuracy is evaluated via standard metrics: MAPE = (100%/N) $\sum |(y_i - y\blacksquare_i)/y_i|$ and $R^2 = 1 - [\sum(y - y\blacksquare)^2 / \sum(y - y\blacksquare)^2]$.

B. Dynamic Pricing Elasticity Multiplier

Optimal shelf price p^* is derived from baseline price p_{base} using a bounded Ridge multiplier $\gamma(\cdot)$:

$$p^* = p_{base} \cdot clip(\gamma(S_r, C_r, D_n, A_r, H_p), 0.75, 1.45)$$

where $S_r = \text{Stock}/\text{MinStock}$, $C_r = P_{comp}/p_{base}$, $D_n = \text{Demand}/100$, A_r is cart abandonment probability, and $H_p \in \{0, 1\}$ is peak hour. Hard clipping prevents predatory pricing or loss-making sales.

C. Autonomous Inventory Control

Replenishment parameters under stochastic daily demand (mean $\bar{d}$, std dev σ_d) and lead time L are defined as:

- **Safety Stock:** $SS = Z_\alpha \cdot \sigma_d \cdot \sqrt{L}$ ($Z_\alpha = 1.65$ for 95% service level).
- **Reorder Point:** $ROP = (\bar{d} \cdot L) + SS$. If $I_t \le ROP$, a PO is triggered.
- **Economic Order Quantity:** $EOQ = \sqrt{(2DS)/H}$.
- **ABC Stratification:** Dynamic 80/15/5 revenue Pareto tiering.

D. Explainable AI (XAI) Logic

Every decision tuple is translated into natural language: If $S_r \le 0.6$ and $D_n \ge 0.70$, the engine issues: *'Surge Pricing: Inventory is below safety threshold while demand remains elevated; price raised by 6.2% to protect stock and secure 38.4% margin.'*

V. IMPLEMENTATION DETAILS

• **Backend:** Python 3.12 Flask REST API, Scikit-learn, NumPy, Pandas. Transactional persistence uses SQLite in Write-Ahead Logging mode (PRAGMA journal_mode=WAL), enabling concurrent non-blocking reads during model retraining.
• **Frontend:** React 18, Vite 6, Tailwind CSS, Recharts for responsive SVG visualization.
• **Governance:** Role-Based Access Control (Admin, Store Manager, Analyst, Supplier) with immutable audit logging.

VI. EXPERIMENTAL RESULTS

Evaluation across 12 product categories over 12 months validated the proposed model suite:

Model	MAE	RMSE	MAPE	R^2
Moving Avg	6.84	9.12	18.4%	0.542
Ridge Reg	4.15	5.88	8.9%	0.761
Random Forest	2.41	3.65	4.8%	0.849
Gradient Boost	**1.72**	**2.48**	**3.2%**	**0.885**

Table II: Forecasting Error Comparison.

Key Outcomes:
• Gradient Boosting achieved **96.8% accuracy (3.2% MAPE)** with training time <150 ms.
• Dynamic pricing achieved **+11.4% gross profit lift**.
• Autonomous ROP triggers reduced stockouts by **37.8%** and holding costs by **22.3%**.

VII. VTU SEMINAR DEFENSE HIGHLIGHTS

• **Why Gradient Boosting over LSTM?** Tabular retail data is heterogeneous; GBDT achieves superior sample efficiency without GPU reliance.
• **Data Robustness:** Vectorized IQR filtering and synonym token resolution ensure self-healing data ingestion.
• **Enterprise Compliance:** Immutable SQLite audit logs ensure end-to-end operational traceability.

VIII. CONCLUSION & FUTURE SCOPE

NeuroRetail delivers an explainable AI decision support platform integrating demand forecasting, dynamic pricing, and inventory replenishment tailored for VTU technical seminar requirements. Future enhancements include IoT smart shelf feeds, multi-store federated learning, and reinforcement learning for multi-competitor game pricing.

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